

Computational Physics

Week 9a

Optimisation

Optimising Code

“Python isn’t as fast as languages like C or fortran”

- We want to demonstrate that python can be just as fast as these languages in most cases
- We’ll highlight places where python code is often written in a sub-optimal way and talk about how to make it faster
- We’ll demonstrate tools to ‘profile’ python code and figure out where the bottlenecks are

In ‘part-II’ we will present an introduction to Cython (week 11b). Cython can be highly effective in those rare instances where python+numpy code cannot be made optimal...scipy uses this extensively.

Coursework 2 (which should be released tomorrow)

- Optimising code is definitely relevant for coursework 2!
- We have carefully designed coursework 2 such that an unoptimised solution will run in 2-3 hours
- An optimised solution can run in a minute (or less)
- It is an option not to optimise your code, but if you wait 2-3 hours and then find a problem (see the debugging class from last year) it will be frustrating!

Optimisation Rule 101

Don't optimise code that does not need to be optimised!

Today

There are 8 notebooks to read/work through

Notebook 3 gives a long(ish) example showing how to optimise a numerical integration code

Notebook 4 is where you can try optimising code - very relevant for coursework 2!

There are plenty of chances in these notebooks to get to grips with profiling code and understanding where optimisation is key